UNIVERSITI TUNKU ABDUL RAHMAN
LEE KONG CHIAN FACULTY OF ENGINEERING AND SCIENCE
LABORATORY SAFETY RULES

The following rules must be obeyed by all students in the Computer laboratories of the faculty. Willful or repeated inadvertent noncompliance may result in dismissal or suspension from the laboratories.

I. No entry without permission:

i) Outsiders are not allowed to enter the laboratory without permission.

ii) No student is allowed to enter the laboratory unless permission has been given by a laboratory staff or a lecturer.

II. At work in the laboratory:

i) The computer facilities at the laboratory are to be used for learning purposes only.

ii) Wear your I.D. cards at all times in a proper visible position whenever you come into the computer laboratory.

iii) It is compulsory to follow dress code of UTAR
   a) Covered shoes (no slippers or flip flops)
   b) Long pants and tidy attire

iv) Do not eat or drink in the laboratory.

v) Do not tamper with any computers, printers, or any other associated equipment. If there are problems or faults, report immediately to the staff at the laboratory.

vi) Do not remove any equipment or document from the computer laboratories.

vii) Do not alter the default hardware and software setup of the computers and printers in the computer laboratories.

viii) Do not copy, rename, change, examine or delete files or information belonging to any other user of the University.

ix) Do not install any unauthorized or unlicensed software into the PC of computer laboratories.

x) Do not make any noise, chit-chat, hold group discussions or disturb other users at the laboratory in any way.

xi) Silent your hand phone when you are in the computer laboratories.

xii) Any equipment broken or damaged must be reported immediately to the laboratory staff.

III. Before leaving the laboratory:

i) No student is allowed to take away any equipment, tools or other property of the laboratory without permission.

ii) Please check and remember to take all your personal belongings such as handphone, wallet, pendrive, hard disk, book, bottle, sweeter, and etc.

iii) Lab staffs are not responsible for lost of personal belongings.

Universiti Tunku Abdul Rahman

Faculty	**Lee Kong Chian Faculty of Engineering and Science**
Department:	**Department of Electrical and Electronics Engineering**
Unit Code and Name	**UEEN 1043 Object-Oriented Concepts and Programming Techniques**
Experiment No.:	**5 / 7**
Title of Experiment:	**Visualization**
Laboratory Room No. and Name:	**Computer lab**
Experiment Duration (hour):	**3 hours**
Number of Student per Group	**1 student**

Objectives

- To automate plot creation for big data analytics using Python programs.

Equipment and Materials

Item Description	***Item category**	Quantity estimation (e.g. per set/group of student)
Personal computer	E	1 per student
Python interpreter (Anaconda Spyder)	S	1 per student

***Item category**	
SP	Sample or specimen
C	Consumable
CH	Chemical
W	Labware, glassware, tool, and components
E	Equipment
S	Software

Caution Statement (Optional)

FACULTY OF ENGINEERING AND SCIENCE
UNIVERSITI TUNKU ABDUL RAHMAN

UEEN 1043 Object-Oriented Concepts and Programming Techniques

Practical 5: Visualization

Objectives

- To automate plot creation for big data analytics using Python programs.

Equipment

- Personal computer with a Python interpreter (Anaconda Spyder)

Introduction

Launch Anaconda Spyder IDE for Python and write codes for the following.

Exercise

5-1 Glob

To automatically list down all the files in a specific directory.

Download **A5a-inputs.zip** and copy the subfolder A5a-inputs to your working directory. In Spyder, double check the current working directory before you run the piece of program below at the console:

```
import glob

# look for all files under folder
files = glob.glob('A5a-inputs/*.csv')

print(files)
```

You should see the result like below:

['A5a-inputs/cov95.csv', ' A5a-inputs/cov85.csv', ' A5a-inputs/cov60.csv', ' A5a-inputs/cov75.csv']

5-2 Reading A Table of Data using Pandas

Data are often saved using a table-like format in a text file. Instead of using the built-in file *open* function to read and process the data, Pandas offers us an easier and more straightforward way to read the table-like data into the programming space with highly readable organization.

```
import pandas as pd

df = pd.DataFrame([[5.1, 3.5, 0], [4.9, 3.0, 0],
                   [7.0, 3.2, 1], [6.4, 3.2, 1],
                   [5.9, 3.0, 2]],
                  columns=['length', 'width', 'species'])
print(df)
```

You should see result like below:

```
   length  width  species
0     5.1    3.5        0
1     4.9    3.0        0
2     7.0    3.2        1
3     6.4    3.2        1
4     5.9    3.0        2
```

To plot using dataframe,

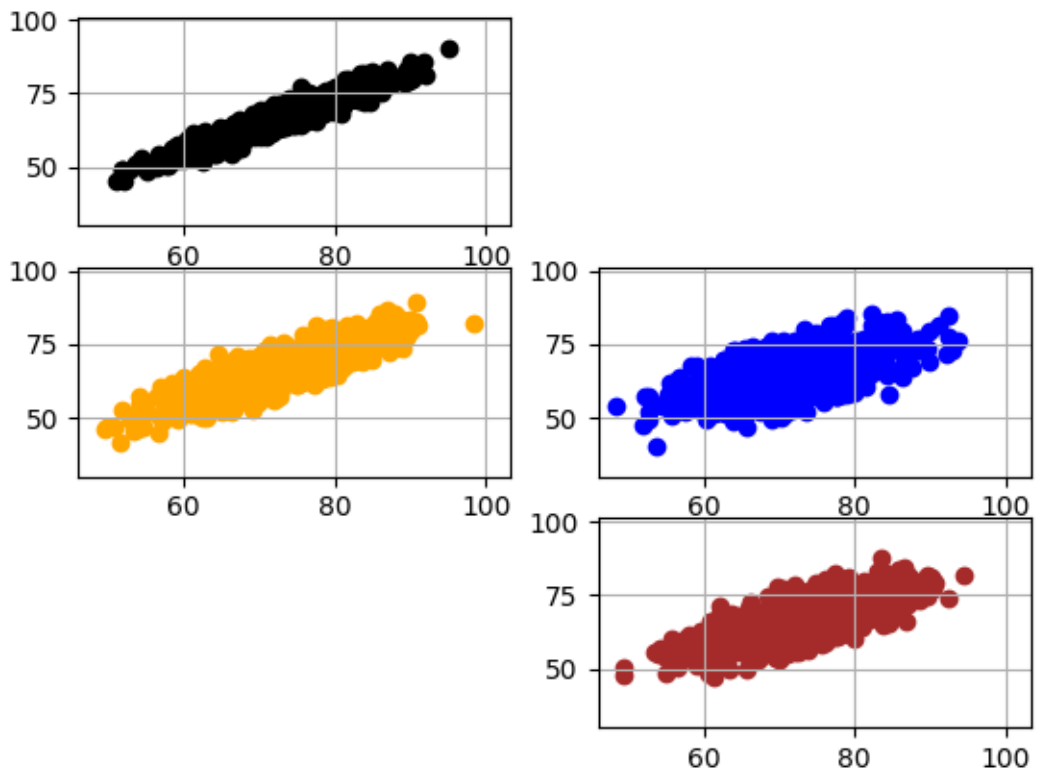
```
import matplotlib.pyplot as plt

ax1 = df.plot.scatter(x='length', y='width', c='red')# dots
ax1.plot(df['length'], df['width'], c='blue')       # lines

plt.show()
```

Exercise

It is a must for you to demonstrate how to look for relevant knowledge/solution on the Internet on your own when you want to find answers beyond the lecture notes in order to complete the following.

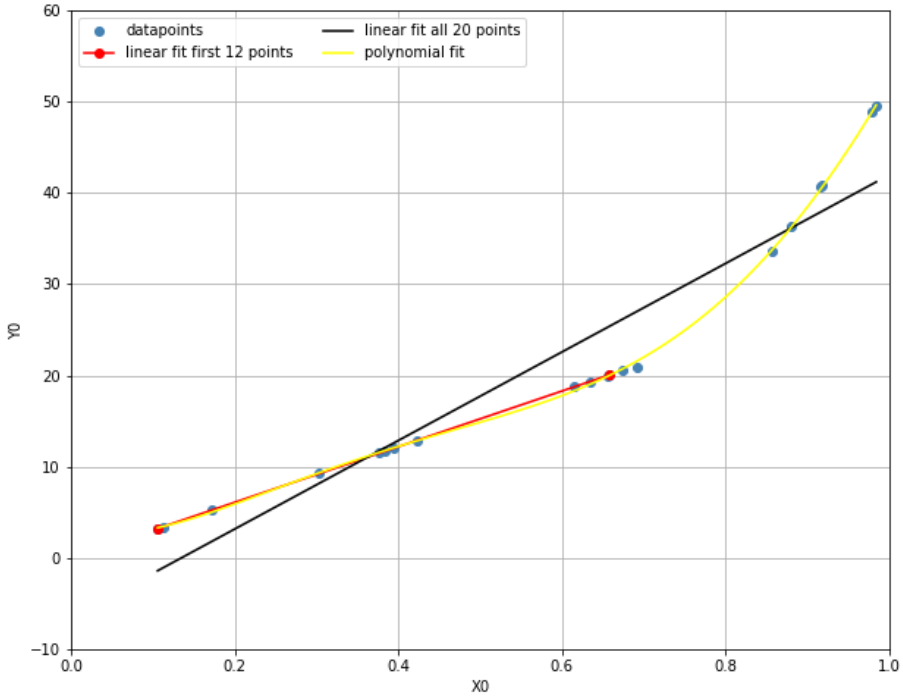
A5a Dictionary creation

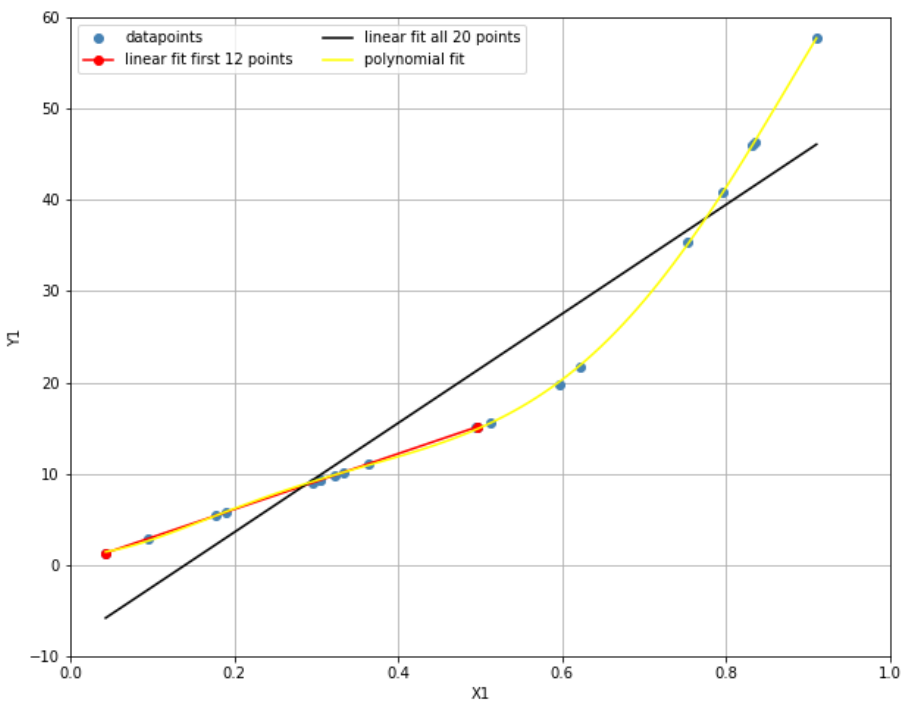
Make use of *glob* module to automatically read the CSV files in **A5a-inputs/** and generate a figure with exactly the same positioning, axes limits, alignments, and colors as shown below based on the data stored in the CSV files. Your solution should be a general one and cannot assume that certain variables are statically assigned.



A5b Big Data Analytics

Refer to ***numpy*** functions ***polyfit()*** and ***poly1d()*** from internet to learn about how to fit the data such that you can generate the figure below based on the input data in **A5b-input.txt**. You are required to fill in the labels for X and Y axes, legends, gap, etc. to match the desired visual effects. The dimension of the figure should be 10x8. The figure below is for your reference. The blue dots are the original data points generated by scatter plot and you are asked to find out the rest.


datapoints
linear fit first 12 points
linear fit all 20 points
polynomial fit
Y0
X0
60
50
40
30
20
10
0
−10
0.0
0.2
0.4
0.6
0.8
1.0


datapoints
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0
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0.0
0.2
0.4
0.6
0.8
1.0